CHENG-YOU HO

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EDUCATION

Yale College: B.S., Computer Science and Mathematics, Certificate in Data Science — Aug 2025 - May 2029

- GPA: 4.00/4.00
- Intermediate Machine Learning, Trustworthy Deep Learning, Algorithmic Game Theory (Fall 2026), Proof-based Linear Algebra, Introduction to Computer Science (Spring 2026), Multivariable Calculus (Fall 2025).
- Graph and Geometric Learning Lab, Splash at Yale, Yale Ballroom Dance Team, Yale Poker Club

National Chiayi Senior High School, Taiwan: Class of Science — Aug 2022 - May 2025

- GPA: 3.96/4.00, SAT: 1550, Valedictorian, Harvard Book Prize, Young Physicists' Tournament Personal Awards

HONORS & AWARDS

- International Physics Olympiad (IPhO) · *Silver Medal* · 2025
- Asian Physics Olympiad (APhO) · *Gold Medal* · 2025
- MIT iQuHack Quantum Computing Hackathon (Alice and Bob) · *1st Place* · 2026
- MOSTRATEC International Science and Technology Fair · *2nd Prize (Representing Taiwan)* · 2024
- International Junior Science Olympiad (IJSO) · *Silver Medal* · 2022

PROFESSIONAL EXPERIENCE

Prof. Rex Ying's Lab *Machine Learning Researcher* — New Haven, Aug 2026 - Present

- Working on knowledge graphs memory systems at the Yale Graph and Geometric Learning Lab.
- Built LLM semantic search pipelines, memory-read mechanisms, and benchmarked retrieval performance.

Banana Labs, Pre-Seed AI NeoLab *Co-founder, RL Environments & Evals* — New York, Jul 2026 - Aug 2026

- Built containerized agentic RL evaluation tasks in numerical reasoning and math proving for $4M pre-seed raise.
- Conducted research on reward signals alongside 6 quantitative co-founders, specializing in high-quality evals.

Yale Quantum Institute *Quantum Machine Learning Researcher* — New Haven, May 2026 - Aug 2026

- Identifying symmetries in quantum ML with LLM-Guided evolutionary search under Professor Steven Girvin.

Sustentra, Pre-Seed AI Fintech Startup *Head of Growth* — New Haven, Apr 2026 - Aug 2026

- Building an AI enterprise product with a Yale SOM '25 founder focused on corporate carbon compliance.
- Designed agentic AI GTM pipeline; financial statements for $2M pre-seed round; website design and tracking.

Yale Undergraduate Capital Partners (KOMPAS VC) *VC Analyst* — New Haven, Oct 2025 - Present

- Conducted due diligence and startup sourcing in climate tech and evaluated emerging investment opportunities.
- Contributed to a memo on presented at the 2025 United Nations Climate Change Conference.

National Taiwan University *Researcher* — Taiwan, Jan 2025 - May 2025

- Analyzed large-scale space-telescope observations (Gaia) using Python to discover new stellar patterns.
- Performed numerical simulations of a galaxy to quantify the effect of dark matter.

LEADERSHIP & COMMUNITY ENGAGEMENT

YQuantum *Co-president* — New Haven, Sep 2026 - Present

- Co-president of Yale's premier quantum computing hackathon, in charge of an organizing committee of 20.

Splash at Yale *Program Director, Incoming Treasurer* — New Haven, 2025 - Present

- Invited high school and middle school students on campus for classes taught by Yale undergraduates.
- Co-organized large-scale events hosting 500+ participants; managed logistics and volunteer coordination.

SKILLS & INTERESTS

Python, PyTorch, TensorFlow, NumPy, pandas, scikit-learn, C, C++, SQL, MySQL, Git, Linux/Unix, Bash/Shell Scripting, Data Structures and Algorithms, Object-Oriented Programming, Software Engineering, Software Testing, Computer Architecture, Systems Design, Distributed Systems, REST APIs, Docker, Machine Learning, Deep Learning, AI Agents, Data Science, Frontend Development, Mandarin Chinese (Native), English (Fluent)